

Momentum-Space QCD Sum-Rule Setup for the $B_c(1P)$ Axial-Vector Mixing Angle

1 Purpose and conventions

We study the mixing between the two axial-vector $B_c(1P)$ basis states associated with the 3P_1 and 1P_1 structures. The current implementation follows the Aliev et al. mixing-angle method, but the OPE is evaluated in momentum space because both quark lines are heavy.

This note records the calculation implemented in `BcMixingMomentum.wl`. It is written so that every convention and intermediate object can be checked independently before using the numerical angle in a paper.

The quark content convention is

$$B_c^+ = \bar{b}c. \quad (1)$$

The two interpolating currents are taken as

$$J_\mu^A = \bar{b} \gamma_\mu \gamma_5 c, \quad (2)$$

$$J_\mu^B = i \bar{b} \sigma_{\mu\alpha} \frac{p^\alpha}{m_b + m_c} \gamma_5 c, \quad (3)$$

where

$$\sigma_{\mu\nu} = \frac{i}{2} [\gamma_\mu, \gamma_\nu]. \quad (4)$$

The explicit factor of i in J_μ^B is retained. This is important because `FeynCalc`'s `DiracSigma` already contains the conventional $i/2$ commutator.

The factor $1/(m_b + m_c)$ is also essential. Since $[\bar{Q}Q] = 3$ and $[p^\alpha] = 1$, the unnormalized tensor current $i\bar{b}\sigma_{\mu\alpha}p^\alpha\gamma_5c$ has mass dimension 4, while J_μ^A has mass dimension 3. The normalized current above makes both currents dimension 3.

If the opposite flavor convention is used, for example $B_c^- = \bar{c}b$, the sign of the quoted angle can change. The absolute convention must therefore be kept fixed when comparing to other papers.

2 Mass-dimension check

The quark field has dimension $3/2$, while p^α , m_b , and m_c have dimension 1. Therefore

$$[J_\mu^A] = 3, \quad [J_\mu^B] = 3 \quad (5)$$

only after the tensor current is normalized by $m_b + m_c$.

For currents with dimensions d_i and d_j ,

$$[\Pi_{\mu\nu}^{ij}] = d_i + d_j - 4, \quad (6)$$

because of the d^4x integration in the two-point function. Thus the normalized basis gives

$$[\Pi_1^{AA}] = [\Pi_1^{AB}] = [\Pi_1^{BB}] = 2. \quad (7)$$

The spectral densities have the same mass dimension as the invariant correlator,

$$[\rho^{AA}] = [\rho^{AB}] = [\rho^{BB}] = 2. \quad (8)$$

In the present code the Borel moment is evaluated as $\int ds e^{-s/M^2} \rho(s)$, so all three numerical moments carry one common extra factor of $[ds] = 2$. This common convention does not affect the mixing angle.

If the tensor current were left unnormalized, the dimensions would instead be

$$[\rho^{AA}], [\rho^{AB}], [\rho^{BB}] = 2, 3, 4, \quad (9)$$

and the matrix entries in the angle formula would not be dimensionally compatible.

3 Implementation map

The main code objects are:

- `Correlator[channel, order]`: $\Pi_1^{ij}(p^2)$ after FeynCalc Dirac algebra and spin projection.
- `SpectralDensity[channel, ‘‘pert’’, s]`: $\rho_{\text{pert}}^{ij}(s)$.
- `NumericBorelPi[channel, order, M2, s0]`: $\Pi_{\text{order}}^{ij}(M^2, s_0)$.
- `NumericMixingAngleDegrees[M2, s0, ‘‘total’’]`: $\theta(M^2, s_0)$ from $\Pi_{\text{pert}}^{ij} + \Pi_{G^2}^{ij} + \Pi_{G^3}^{ij}$.

The implemented OPE orders are

$$\text{pert}, \quad G^2, \quad \text{pertG2} = \text{pert} + G^2, \quad G^3, \quad \text{total} = \text{pert} + G^2 + G^3. \quad (10)$$

The implemented dimension-6 triple-gluon condensate $\langle g_s^3 G^3 \rangle$ is the standard vacuum-averaged single-line propagator term. Possible cross-line open-field G^3 terms need a separate derivation.

4 Correlators and spin-1 projection

Define

$$\Pi_{\mu\nu}^{ij}(p) = i \int d^4x e^{ip \cdot x} \langle 0 | T \{ J_\mu^i(x) J_\nu^{j\dagger}(0) \} | 0 \rangle, \quad i, j \in \{A, B\}. \quad (11)$$

The invariant amplitude used for the axial-vector spin-1 state is extracted with

$$P_{(1)}^{\mu\nu} = \frac{1}{3} \left(g^{\mu\nu} - \frac{p^\mu p^\nu}{p^2} \right), \quad (12)$$

so that

$$\Pi_1^{ij}(p^2) = P_{(1)}^{\mu\nu} \Pi_{\mu\nu}^{ij}(p). \quad (13)$$

The mixing angle is then obtained from

$$\tan 2\theta = -\frac{2\Pi_1^{AB}}{\Pi_1^{AA} - \Pi_1^{BB}}. \quad (14)$$

Numerically we use the quadrant-safe definition

$$\theta_{\text{raw}} = \frac{1}{2} \text{atan2}(-2\Pi_1^{AB}, \Pi_1^{AA} - \Pi_1^{BB}), \quad (15)$$

and quote the principal branch

$$\theta = \theta_{\text{raw}} - \frac{\pi}{2} \text{Round} \left(\frac{\theta_{\text{raw}}}{\pi/2} \right), \quad \theta \in [-45^\circ, 45^\circ]. \quad (16)$$

5 Momentum-space OPE

The perturbative heavy-quark propagator is

$$S_Q^{(0)}(k) = \frac{\not{k} + m_Q}{k^2 - m_Q^2}. \quad (17)$$

For the dimension-4 gluon condensate we use the background-field expansion

$$S_Q(k) = S_Q^{(0)}(k) + g_s G_A^{\alpha\beta} T^A S_{Q,\alpha\beta}^{(G)}(k) + S_Q^{(G^2)}(k), \quad (18)$$

where

$$S_{Q,\alpha\beta}^{(G)}(k) = -\frac{1}{4} \frac{\sigma_{\alpha\beta}(\not{k} + m_Q) + (\not{k} + m_Q)\sigma_{\alpha\beta}}{(k^2 - m_Q^2)^2}, \quad (19)$$

$$S_Q^{(G^2)}(k) = \frac{\langle g_s^2 G^2 \rangle}{12} m_Q \frac{k^2 + m_Q}{(k^2 - m_Q^2)^4}. \quad (20)$$

Equivalently,

$$g_s G_A^{\alpha\beta} T^A S_{Q,\alpha\beta}^{(G)}(k) = -\frac{g_s G_A^{\alpha\beta} T^A}{4} \frac{\sigma_{\alpha\beta}(\not{k} + m_Q) + (\not{k} + m_Q)\sigma_{\alpha\beta}}{(k^2 - m_Q^2)^2}. \quad (21)$$

The one-gluon fields are contracted with

$$\langle g_s^2 G_A^{\alpha\beta} G_B^{\rho\sigma} \rangle = \frac{\delta^{AB}}{96} \langle g_s^2 G^2 \rangle \left(g^{\alpha\rho} g^{\beta\sigma} - g^{\alpha\sigma} g^{\beta\rho} \right), \quad (22)$$

and

$$\text{Tr}(T^A T^B) = \frac{\delta^{AB}}{2}. \quad (23)$$

Thus the color factor for the $S^{(G)}S^{(G)}$ piece is

$$\frac{N_c^2 - 1}{192} \langle g_s^2 G^2 \rangle, \quad N_c = 3. \quad (24)$$

The three projected OPE contributions are generated from

$$\Pi_1^{AA} \sim P_{(1)}^{\mu\nu} \text{Tr} [\gamma_\mu \gamma_5 S_c(k) \gamma_\nu \gamma_5 S_b(k-p)], \quad (25)$$

$$\Pi_1^{AB} \sim P_{(1)}^{\mu\nu} \text{Tr} \left[\gamma_\mu \gamma_5 S_c(k) i\sigma_{\nu\alpha} \frac{p^\alpha}{m_b + m_c} \gamma_5 S_b(k-p) \right], \quad (26)$$

$$\Pi_1^{BB} \sim P_{(1)}^{\mu\nu} \text{Tr} \left[i\sigma_{\mu\alpha} \frac{p^\alpha}{m_b + m_c} \gamma_5 S_c(k) i\sigma_{\nu\beta} \frac{p^\beta}{m_b + m_c} \gamma_5 S_b(k-p) \right]. \quad (27)$$

6 Perturbative spectral densities

For the perturbative loop, impose the two-particle cut

$$k^2 = m_c^2, \quad (k-p)^2 = m_b^2, \quad (28)$$

which gives

$$k \cdot p = \frac{s + m_c^2 - m_b^2}{2}, \quad s = p^2. \quad (29)$$

Define

$$\lambda(s, m_b, m_c) = s^2 + m_b^4 + m_c^4 - 2sm_b^2 - 2sm_c^2 - 2m_b^2 m_c^2. \quad (30)$$

The perturbative spectral densities are implemented as

$$\rho_{\text{pert}}^{ij}(s) = \frac{N_c}{16\pi^2} \frac{\sqrt{\lambda(s, m_b, m_c)}}{s} \frac{N^{ij}(s)}{(m_b + m_c)^{n_B^{ij}}}, \quad (31)$$

where $n_B^{AA} = 0$, $n_B^{AB} = n_B^{BA} = 1$, and $n_B^{BB} = 2$ counts how many tensor currents occur in the channel. The unnormalized numerator functions are

$$N^{AA}(s) = -\frac{2}{s} [(m_b + m_c)^2 - s] [(m_b - m_c)^2 + 2s], \quad (32)$$

$$N^{AB}(s) = 6(m_b - m_c) [(m_b + m_c)^2 - s], \quad (33)$$

$$N^{BB}(s) = 2 [-2(m_b^2 - m_c^2)^2 + (m_b^2 - 6m_b m_c + m_c^2)s + s^2]. \quad (34)$$

Equivalently, after setting $N_c = 3$,

$$\rho_{\text{pert}}^{AA}(s) = -\frac{3}{8\pi^2 s^2} [(m_b + m_c)^2 - s] [(m_b - m_c)^2 + 2s] \sqrt{\lambda(s, m_b, m_c)}, \quad (35)$$

$$\rho_{\text{pert}}^{AB}(s) = \frac{9}{8\pi^2 s(m_b + m_c)} (m_b - m_c) [(m_b + m_c)^2 - s] \sqrt{\lambda(s, m_b, m_c)}, \quad (36)$$

$$\rho_{\text{pert}}^{BB}(s) = \frac{3}{8\pi^2 s(m_b + m_c)^2} [-2(m_b^2 - m_c^2)^2 + (m_b^2 - 6m_b m_c + m_c^2)s + s^2] \sqrt{\lambda(s, m_b, m_c)}. \quad (37)$$

The perturbative Borel moments are

$$\Pi_{\text{pert}}^{ij}(M^2, s_0) = \int_{(m_b + m_c)^2}^{s_0} ds e^{-s/M^2} \rho_{\text{pert}}^{ij}(s). \quad (38)$$

With this convention all three perturbative spectral densities have mass dimension 2. Without the factors of $(m_b + m_c)^{-n_B^{ij}}$, their dimensions would be 2, 3, and 4 for AA , AB , and BB , respectively.

7 Dimension-4 and dimension-6 gluon condensates

The implemented condensate contribution is

$$\Pi_{G^2}^{ij} = \Pi^{ij}[S_c^{(G^2)} S_b^{(0)}] + \Pi^{ij}[S_c^{(0)} S_b^{(G^2)}] + \Pi^{ij}[S_c^{(G)} S_b^{(G)}]. \quad (39)$$

After Feynman parametrization the terms have the schematic form

$$\mathcal{A}_{G^2}^{ij}(x_1, x_2, s) = \sum_r C_r^{ij}(x_1, x_2, s) [m_c^2 x_1^2 + (m_b^2 + m_c^2 - s)x_1 x_2 + m_b^2 x_2^2]^{-n_r}. \quad (40)$$

The code then fixes the simplex by

$$x_1 = x, \quad x_2 = 1 - x. \quad (41)$$

This gives

$$\Delta(x, s) = m_c^2 x + m_b^2 (1 - x) - s x (1 - x). \quad (42)$$

For the triple-gluon condensate we use the standard momentum-space vacuum-averaged heavy-quark propagator term

$$S_Q^{(G^3)}(k) = \frac{\langle g_s^3 G^3 \rangle}{48} \frac{(\not{k} + m_Q) [\not{k}(k^2 - 3m_Q^2) + 2m_Q(2k^2 - m_Q^2)] (\not{k} + m_Q)}{(k^2 - m_Q^2)^6}. \quad (43)$$

The implemented dimension-6 contribution is

$$\Pi_{G^3}^{ij} = \Pi^{ij}[S_c^{(G^3)} S_b^{(0)}] + \Pi^{ij}[S_c^{(0)} S_b^{(G^3)}]. \quad (44)$$

This is the standard single-heavy-propagator contribution. It does not yet include any possible cross-line open-field terms of the schematic type $S^{(G_{\text{open}}^2)} S^{(G)}$, which require an explicitly open two-gluon propagator and a separate color/Lorentz derivation.

The G^2 and G^3 terms are not treated as ordinary smooth spectral densities. We instead use a direct Borel transform. Set $s = -Q^2$ and write

$$\Delta(x, -Q^2) = x(1-x) [Q^2 + \bar{s}(x)], \quad (45)$$

where

$$\bar{s}(x) = \frac{m_c^2 x + m_b^2 (1-x)}{x(1-x)}. \quad (46)$$

The Feynman-parameter amplitude is decomposed as

$$-i \mathcal{A}_{Gr}^{ij}(x, -Q^2) = \sum_{\ell \geq 1} \frac{a_\ell^{ij}(x)}{[Q^2 + \bar{s}(x)]^\ell} + \text{polynomial subtraction terms}. \quad (47)$$

The polynomial subtraction terms vanish after the Borel transform. The working phase convention is the explicit factor $-i$ shown above. This is a bookkeeping convention for removing the loop-integration prefactor returned by FeynCalc's Feynman-parameter amplitude. This is the convention that should be independently checked.

We use

$$\mathcal{B}_{M^2} \left[\frac{1}{(Q^2 + \bar{s})^n} \right] = \frac{1}{(n-1)! (M^2)^{n-1}} e^{-\bar{s}/M^2}. \quad (48)$$

Thus

$$\Pi_{Gr}^{ij}(M^2, s_0) = \int_{x_-}^{x_+} dx e^{-\bar{s}(x)/M^2} \sum_{\ell \geq 1} \frac{a_\ell^{ij}(x)}{(\ell-1)! (M^2)^{\ell-1}}. \quad (49)$$

The continuum restriction is

$$\bar{s}(x) \leq s_0, \quad (50)$$

which gives

$$x_\pm = \frac{s_0 + m_b^2 - m_c^2 \pm \sqrt{\lambda(s_0, m_b, m_c)}}{2s_0}. \quad (51)$$

8 Total Borel moments and numerical test point

The total moments used in the current numerical result are

$$\Pi^{ij}(M^2, s_0) = \Pi_{\text{pert}}^{ij}(M^2, s_0) + \Pi_{G^2}^{ij}(M^2, s_0) + \Pi_{G^3}^{ij}(M^2, s_0). \quad (52)$$

The current default numerical inputs are

$$m_b = 4.18 \text{ GeV}, \quad (53)$$

$$m_c = 1.27 \text{ GeV}, \quad (54)$$

$$\langle g_s^2 G^2 \rangle = 4\pi^2 (0.012) \text{ GeV}^4, \quad (55)$$

$$\langle g_s^3 G^3 \rangle = 0.57 \text{ GeV}^6, \quad (56)$$

$$M^2 = 10 \text{ GeV}^2, \quad (57)$$

$$s_0 = 55 \text{ GeV}^2. \quad (58)$$

For this point,

$$x_- = 0.3322728664, \quad x_+ = 0.9560816791. \quad (59)$$

The perturbative moments are

$$\Pi_{\text{pert}}^{AA} = 0.1441716826, \quad (60)$$

$$\Pi_{\text{pert}}^{AB} = -0.1051577192, \quad (61)$$

$$\Pi_{\text{pert}}^{BB} = 0.1348134513. \quad (62)$$

The gluon-condensate moments are

$$\Pi_{G^2}^{AA} = -0.0003658277, \quad (63)$$

$$\Pi_{G^2}^{AB} = 0.0002309420, \quad (64)$$

$$\Pi_{G^2}^{BB} = -0.0001888869. \quad (65)$$

The triple-gluon-condensate moments are

$$\Pi_{G^3}^{AA} = 0.0000700979, \quad (66)$$

$$\Pi_{G^3}^{AB} = -0.0000493659, \quad (67)$$

$$\Pi_{G^3}^{BB} = 0.0000310238. \quad (68)$$

Thus

$$\Pi_{\text{total}}^{AA} = 0.1438759528, \quad (69)$$

$$\Pi_{\text{total}}^{AB} = -0.1049761432, \quad (70)$$

$$\Pi_{\text{total}}^{BB} = 0.1346555882. \quad (71)$$

The corresponding angles are

$$\theta_{\text{pert}} = 43.726118943^\circ, \quad (72)$$

$$\theta_{\text{pert}+G^2} = 43.747426660^\circ, \quad (73)$$

$$\theta_{\text{pert}+G^2+G^3} = 43.742693524^\circ. \quad (74)$$

At this point the relative condensate corrections are

$$\frac{\Pi_{G^2}^{AA}}{\Pi_{\text{pert}}^{AA}} = -0.002537, \quad (75)$$

$$\frac{\Pi_{G^2}^{AB}}{\Pi_{\text{pert}}^{AB}} = -0.002196, \quad (76)$$

$$\frac{\Pi_{G^2}^{BB}}{\Pi_{\text{pert}}^{BB}} = -0.001401, \quad (77)$$

$$\frac{\Pi_{G^3}^{AA}}{\Pi_{\text{pert}}^{AA}} = 0.000486, \quad (78)$$

$$\frac{\Pi_{G^3}^{AB}}{\Pi_{\text{pert}}^{AB}} = 0.000469, \quad (79)$$

$$\frac{\Pi_{G^3}^{BB}}{\Pi_{\text{pert}}^{BB}} = 0.000230. \quad (80)$$

9 Perturbative α_s sensitivity model

The full perturbative $O(\alpha_s)$ correction is not yet calculated. As a diagnostic, the code implements a channel-dependent trial rescaling of the leading perturbative moment,

$$\Pi_{\text{pert}}^{ij}(M^2, s_0) \longrightarrow \left[1 + \frac{\alpha_s}{\pi} K_{ij}\right] \Pi_{\text{pert}}^{ij}(M^2, s_0). \quad (81)$$

The condensate pieces are left unchanged, so for the current total moment

$$\Pi_K^{ij} = \left[1 + \frac{\alpha_s}{\pi} K_{ij}\right] \Pi_{\text{pert}}^{ij} + \Pi_{G^2}^{ij} + \Pi_{G^3}^{ij}. \quad (82)$$

This is only a sensitivity model; it is not a substitute for the real two-loop NLO perturbative spectral density.

For example, with $\alpha_s = 0.26$ and

$$K_{AA} = 1, \quad K_{AB} = 0, \quad K_{BB} = -1, \quad (83)$$

the total angle at $M^2 = 10 \text{ GeV}^2$, $s_0 = 55 \text{ GeV}^2$ changes from

$$\theta_{\text{base}} = 43.742693524^\circ \quad (84)$$

to

$$\theta_K = 40.625721673^\circ, \quad \Delta\theta = -3.116971851^\circ. \quad (85)$$

A coarse scan with $K_{AA}, K_{AB}, K_{BB} \in \{-2, 0, 2\}$ gives a maximum branch-corrected shift of about

$$\max |\Delta\theta| \simeq 7.52^\circ. \quad (86)$$

10 How to reproduce the numerical check

In Mathematica, load the implementation with

```
SetDirectory["/Users/sbilmis/Bc_mixing"];
Get["BcMixingMomentum.wl"];
```

Then evaluate

```
NumericOPESummary[10, 55]
NumericMixingAngleDegrees[10, 55, "pert"]
NumericMixingAngleDegrees[10, 55, "pertG2"]
NumericMixingAngleDegrees[10, 55, "total"]
KFactorSensitivityRecord[10, 55, KFactorAssociation[1, 0, -1], 0.26, "total"]
KFactorSensitivityEnvelope[10, 55, {-2, 2, 2}, 0.26, "total"]
MixingMassDimensionReport[]
CheckMixingMassDimensions[]
PerturbativeSpectralDensityDimensionReport[]
```

For window studies one can scan

```
MixingAngleTable[{8, 12, 1}, {50, 60, 5}, "total"]
OPEConvergenceDataset[{8, 12, 1}, {50, 60, 5}]
```

11 Checklist for independent verification

A theorist can check the calculation in the following order:

1. Confirm the current convention, especially the explicit factor of i in J_μ^B , the normalization $p^\alpha/(m_b + m_c)$, and the definition of $\sigma_{\mu\nu}$.
2. Reproduce the three projected perturbative traces and verify the numerator functions $N^{AA}(s)$, $N^{AB}(s)$, and $N^{BB}(s)$.
3. Check the two-body phase-space normalization $N_c\sqrt{\lambda}/(16\pi^2s)$ used in $\rho_{\text{pert}}^{ij}(s)$.
4. Verify the background-field propagator terms and the color factor in the $S^{(G)}S^{(G)}$ contribution.
5. Independently derive the direct Borel formula for the G^2 and G^3 terms, including the $-i$ phase convention.
6. Check whether the standard single-line G^3 propagator contribution is sufficient, or whether cross-line open-field G^3 terms must be added.
7. Check that $\Pi_{G^2}/\Pi_{\text{pert}}$ and $\Pi_{G^3}/\Pi_{\text{pert}}$ remain small in the accepted Borel window.
8. Use the K_{ij} sensitivity model only as an estimate of possible channel-dependent perturbative $O(\alpha_s)$ uncertainty.
9. Derive the true $O(\alpha_s)$ two-loop perturbative spectral densities separately.
10. Determine the final working window using OPE convergence, Borel stability, and pole-dominance criteria.